AUTUMN INTERNSHIP PROJECT REPORT

ACADEMIC SUCCESS FORECASTING

Submitted by:

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1. Abstract

This project forecasts the academic success of secondary-school students from demographic, family and lifestyle data, and groups them into learner segments. Using the 395-record *student_mat.csv* dataset [1], a new feature named Total_Grade (G1 + G2 + G3) was engineered as the target variable, and dropped the three original grade columns to prevent data leakage. Correlation analysis, one-hot encoding, standard scaling and Principal Component Analysis (PCA) prepared the data for K-Means clustering, which separated students into three profiles, from high achievers to an at-risk group with the lowest study time and the most absences.

A baseline Linear Regression (RMSE 10.12, R^2 0.23) outperformed a grid-search-tuned MLP Regressor (RMSE 10.68, R^2 0.14) on the 20% hold-out set, which suggests that a simple linear model generalizes better on a dataset this small. The report closes with a recommendation to flag students with past failures or low study time early and offer them specific tutoring sessions.

2. Introduction

Schools can only help struggling students if they spot them early, and this project tests how far background and behavior data can support that. This project was completed as a part of Autumn Internship Program 2026 at IDEAS (Institute of Data Engineering, Analytics and Science Foundation), ISI Kolkata under the guidance of Dr. Chandan Biswas.

The work follows a standard data-science pipeline: cleaning, target engineering, exploratory analysis, preprocessing, unsupervised segmentation and supervised regression. The code is written in Python with pandas, seaborn and scikit-learn.

3. Project Objective

1. Load and clean the student performance data using pandas and build a target variable Total_Grade without leaking the grade information into the features.
2. Find the features most strongly correlated with Total_Grade and explain why.
3. Segment students with PCA and K-Means and describe the academic profile of each segment.
4. Compare a baseline Linear Regression with a cross-validated MLP Regressor using RMSE and R^2 score.
5. Recommend one data-backed intervention to reduce student failure.

4. Methodology

The analysis ran in ten steps, each kept simple and commented in the notebook.

1. **Data loading**. The CSV was read into a pandas DataFrame with pd.read_csv, and its shape (395, 33) and first three rows were checked.
2. **Cleaning**. Fully duplicated rows were removed with drop_duplicates and missing values with dropna. Duplicates would give some students extra weight, and dropping beats imputing here because a few lost rows cost less than invented values. In this dataset neither problem occurred, so the shape stayed (395, 33).
3. **Target engineering**. Total_Grade = G1 + G2 + G3 was created, and G1, G2 and G3 were dropped, because the target is a direct function of them and keeping them would leak the answer into the features.
4. **Correlation analysis**. The 14 numerical columns were correlated and drawn as a seaborn heatmap.
5. **Encoding**. Features (X) were separated from the target (y), and encode 17 categorical columns by one-hot encoding with drop_first=True, giving 39 numerical features.
6. **Scaling**. StandardScaler gave every feature mean 0 and standard deviation 1, and the result was turned back into a DataFrame with the original column names.
7. **PCA**. The scaled data was reduced to two components, PC1 and PC2, stored in pca_df.
8. **Clustering**. K-Means (random_state=42, n_init=10) was run for k = 1 to 10 on pca_df to plot inertia, then fitted with k = 3 and the labels were added to the unscaled data.
9. **Splitting and modelling**. The scaled features and target were split 80/20 with random_state=42, giving 316 training and 79 test records. A baseline Linear Regressor and an MLP Regressor with one hidden layer of 16 neurons (max_iter=1000) were trained.
10. **Tuning and evaluation**. GridSearchCV with 3-fold cross-validation searched hidden_layer_sizes of (16,), (32,) and (16, 8) against alpha of 0.0001, 0.01 and 1.0, which is nine settings and 27 fits. Every model was scored on the test set with RMSE and R^2.

5. Data Analysis and Results

5.1 Correlations with Total_Grade

Past class failures are the strongest correlate of Total_Grade (r = -0.38), followed by mother's education, Medu (r = +0.22). The heatmap below shows the complete correlation of the all 14 numerical columns.

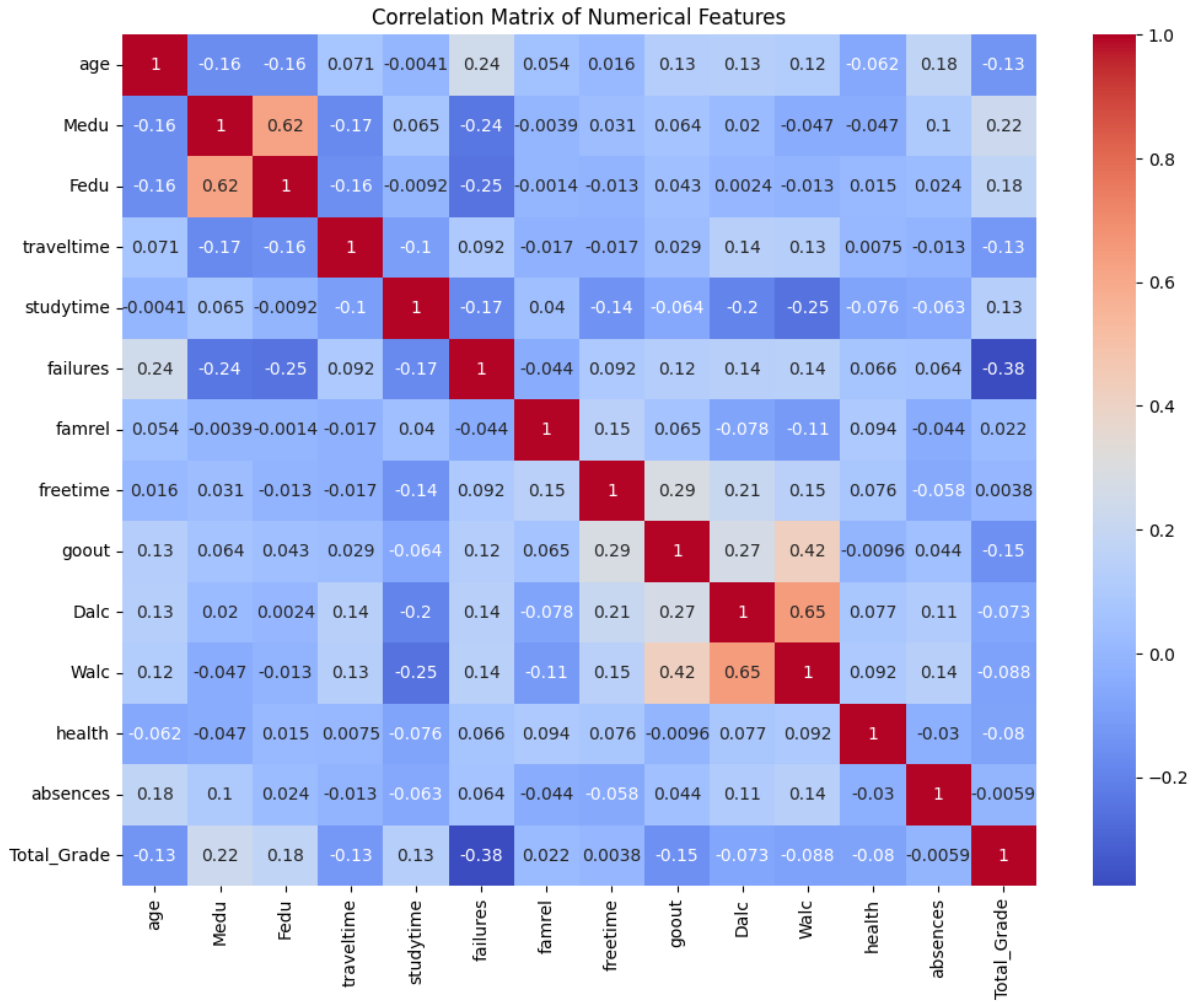


Figure 1: Correlation heatmap of numerical features.

Students with more past failures tend to have weaker foundations, so their grades are lower. A better-educated mother usually means more support with schoolwork at home, which results a higher grade. Father's education (+0.18) and weekly study time (+0.14) point the same way, while going out with friends (-0.16) and age (-0.14) leads to lower grades. Absences show almost no linear link (r = -0.006). No single feature passes |r| = 0.38, so the grades depend on many small factors, which limits how well any model can predict them.

5.2 Learner segments

The elbow curve falls smoothly from an inertia of 2,383 at k = 1 to 321 at k = 10, with no sharp bends. The gain from each extra cluster shrinks from 514 (k = 2 to 3) to 240 (k = 3 to 4), so k = 3 was used, as the brief specified.

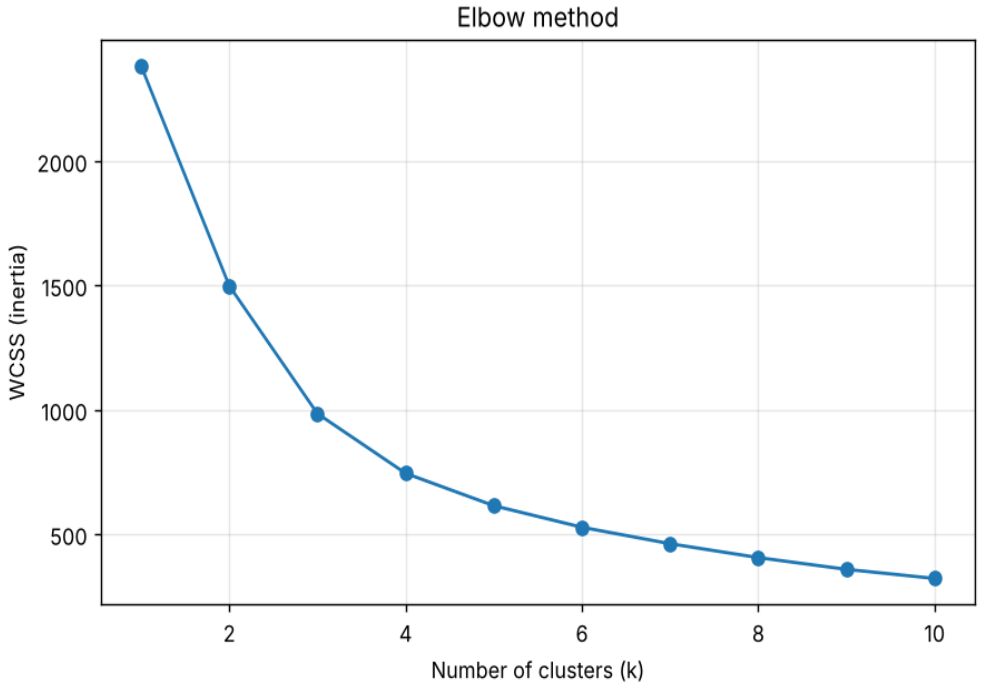


Figure 2: Elbow method for determining the optimal number of clusters.

The three K-Means clusters, fitted on the two principal components and profiled on the original unscaled data, look like this:

Cluster	Students	Mean Total_Grade	Mean study time (1–4)	Mean absences	Profile
0	142	35.68	2.05	5.58	High achievers
1	182	31.04	2.24	5.45	Hardworking average students
2	71	27.32	1.48	6.63	At-risk students

Table 1: Student Segment Profiling by Cluster Means

Cluster 0 has the highest grades with average study time and few absences. Cluster 1 has the most study time and the fewest absences, yet scores in the middle. Cluster 2 has the lowest grades, the least study time and the most absences.

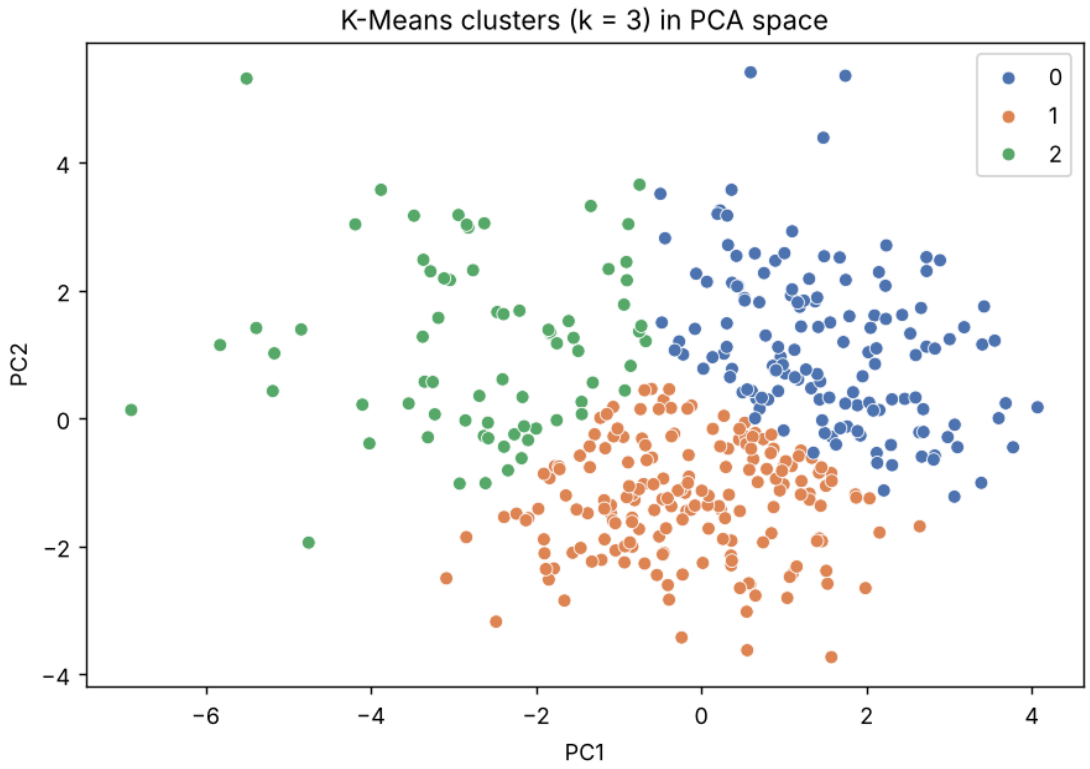


Figure 3: K-Means clusters (k=3) in PCA Space

5.3 Modelling and Results

The baseline Linear Regression beat the grid-search-tuned MLP on the 79-student test set, with an RMSE of 10.12 against 10.68 and an R^2 of 0.23 against 0.14.

Model	RMSE (grade points)	R^2
Linear Regression (baseline)	10.12	0.23
MLP Regressor (untuned)	10.38	0.19
MLP Regressor, (Tuned)	10.68	0.14

Table 2: Different model comparison

The grid search chose a single hidden layer of 32 neurons and alpha = 1.0, yet its best cross-validated R^2 on the training data was -0.23. This shows the network predicted worse than simply using the average grade, and the test result (R^2: 0.14) confirms it. The untuned MLP Regressor (RMSE: 10.38, R^2: 0.19) actually did better than the tuned MLP. That result and the cross-validation score come from a supplementary check on the same train/test split, since the 15 questions do not ask for them.

For scale, the test grades have a standard deviation of 11.62, so even the best model (RMSE 10.12) only modestly beats guessing the mean. Two things explain the weak scores. The strongest predictors, the interim grades G1 and G2, were deliberately removed, and with only 316 training records the neural network has too little data to learn stable patterns.

6. Conclusion

Past failures, not absences, are the clearest warning sign of low grades, and a simple Linear Regression predicted grades better than a tuned neural network on this data.

The baseline gave an R^2 of 0.23 and an RMSE of 10.12, while the tuned MLP gave 0.14 and 10.68. With only 395 records, the extra flexibility of the network brought overfitting rather than accuracy. K-Means found three learner segments, and the at-risk segment holds 71 students, about 18% of the cohort.

Recommendations.

The institution should run an early-warning tutoring program. At the start of each term it would flag students with one or more past failures, or with low study time, and give them mandatory tutoring and study-skills support. This targets the strongest correlate of low grades (failures, $r = -0.38$) and matches the at-risk profiles which has the lowest study time(1.48). These are associations in the data, so the program's effect should be tested rather than assumed.

Limitations

- PCA keeps only 15.5% of the variance (8.7% + 6.8%), so the two-dimensional clusters are a coarse view of the students.
- The elbow curve has no sharp bend, and $k = 3$ was taken from the brief.
- Features were scaled on all 395 records before the 80/20 split, as the brief ordered. Fitting the scaler on training data only would be stricter.
- Results come from one split with 79 test students and would shift with a different random state.
- The interim grades were removed by design, which caps how well any model can score.

Future work

- Try Ridge, Lasso, Random Forest and Gradient Boosting, and compare them with repeated cross-validation instead of a single split.
- Include G1 back as a feature to build a realistic mid-term early-warning model.
- Collect more records, since the dataset is too small for neural networks to shine.

Tools and References

Tools. Python with pandas and NumPy for data handling, matplotlib and seaborn for plots, and scikit-learn for StandardScaler, PCA, KMeans, train_test_split, GridSearchCV, LinearRegression, MLPRegressor and the RMSE and R^2 metrics.

7. APPENDICES

The following supporting material accompanies this report.

7.1. References:

[1] Student performance dataset (student-mat.csv), used in this project: https://raw.githubusercontent.com/guipsamora/pandas_exercises/master/04_Apply/Students_Alcohol_Consumption/student-mat.csv

[2] Cortez, P. and Silva, A. (2008). Using Data Mining to Predict Secondary School Student Performance. Proceedings of the 5th Future Business Technology Conference (FUBUTEC 2008), pp. 5-12, Porto, Portugal.
Dataset page: https://archive.ics.uci.edu/dataset/320/student+performance

[3] scikit-learn documentation (StandardScaler, PCA, KMeans, GridSearchCV, LinearRegression, MLPRegressor): https://scikit-learn.org/stable/

7.2. Survey Questionnaire: Not applicable. No survey was conducted; the project uses a publicly available secondary dataset.

7.3. Other documents: Google Colab notebook with all 15 questions and answers: https://colab.research.google.com/drive/1ra6gowPh4j4BiVaPt2S8e3cZlWRTxGS0?usp=sharing ;
Copy of this report: https://github.com/arun-sankar03/ISI-Autumn-Internship-2026-Academic-Success-Forecasting